

Linear and Discrete Optimization Modelling

UNSW MATH3171 25T3

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Chapter 1

Introduction to Optimization

1.1 Optimization: An Introduction

Many problems we counter in a large number of area entail trying to achieve the “best” outcome subject to some “restrictions”. The mathematical field that encompasses these problems is called Optimization.

An optimization problem consist of three parts: decision variables, object functions and constraints.

- (1) **Decision variables:** what can you change or what you wish to determine.

Denote the decision variable $\mathbf{x} \in H$ where H is a finite dimensional space. For example,

- $H = \mathbb{R}^n$ and the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$, the number of variables n .

– Column vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$;

– Univariate: $n = 1$;

– Multivariate: $n = 2, 3, 4, \dots$ up to millions;

- $H = \mathbb{R}^{p \times n}$ where $\mathbb{R}^{p \times n}$ is the set of $(p \times n)$ matrices, and the decision variable $\mathbf{x} = X \in \mathbb{R}^{p \times n}$ is an $(p \times n)$ matrix.

- (2) **Objective functions:** a mathematical function of the variables quantifying the idea of “best”.

- $H = \mathbb{R}^n$, the decision variable is a vector $\mathbf{x} \in \mathbb{R}^n$ and objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. For example,

– Linear: $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$, $\mathbf{c} \in \mathbb{R}^n$ fixed;

– Affine: $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} + \gamma$, $\mathbf{c} \in \mathbb{R}^n, \gamma \in \mathbb{R}$ fixed;

– Quadratic: $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T G \mathbf{x} + \mathbf{c}^T \mathbf{x} + \gamma$ $G \in \mathbb{R}^{n \times n}, \mathbf{c} \in \mathbb{R}^n, \gamma \in \mathbb{R}$ fixed.

- $H = \mathbb{R}^{p \times n}$, the decision variable is an $(p \times n)$ matrix X and objective function $f : \mathbb{R}^{p \times n} \rightarrow \mathbb{R}$. For example,

– $f(X) = \sum_{i=1}^p \sum_{j=1}^n A_{ij} X_{ij}$, where A is given $(p \times n)$ matrix X ;

– $f(X) = \text{rank}(X)$, where $\text{rank}(X)$ is the rank of a matrix X .

- (3) **Constraints:** describe restrictions on the allowable values of variables.

- $H = \mathbb{R}^n$,
 - Equality constraints $g_i(\mathbf{x}) = 0$, $i = 1, \dots, m_E$ (e.g. binary constraint: $x_i \in \{0, 1\}$);
 - Inequality constraints $g_i(\mathbf{x}) \leq 0$, $i = m_E + 1, \dots, m$;
 - Geometric constraints $\mathbf{x} \in C \subseteq \mathbb{R}^n$;
 - Feasible region $\Omega \in \mathbb{R}^n$

$$\Omega = \left\{ \mathbf{x} \in C : \begin{array}{l} g_i(\mathbf{x}) = 0, i = 1, \dots, m_E; \\ g_i(\mathbf{x}) \leq 0, i = m_E + 1, \dots, m \end{array} \right\}$$

- The case for $H = \mathbb{R}^{p \times n}$ is similar, except the constraint functions g_i are functions on $\mathbb{R}^{p \times n}$ and the set C in the geometric constraints is a set in $\mathbb{R}^{p \times n}$. In this case, the feasible region $\Omega \subset \mathbb{R}^{p \times n}$

$$\Omega = \left\{ X \in C \subseteq \mathbb{R}^{p \times n} : \begin{array}{l} g_i(X) = 0, i = 1, \dots, m_E; \\ g_i(X) \leq 0, i = m_E + 1, \dots, m \end{array} \right\}$$

Optimization Problem A typical optimization problem looks like

$$\begin{array}{ll} \min_{x \in H} & f(\mathbf{x}) \\ \text{such that} & g_i(\mathbf{x}) = 0, i = 1, \dots, m_E, \\ & g_i(\mathbf{x}) \leq 0, i = m_E + 1, \dots, m, \\ & x \in C \subseteq H \end{array}$$

where H is the finite dimensional space.

Linear Programming A linear program is an optimization problem where the underlying space $H = \mathbb{R}^n$, all the functions $f, g_i, i = 1, \dots, m$ are affine functions on $\mathbb{R}^n$ and there is no geometric constraints:

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & \mathbf{c}^T \mathbf{x} + \gamma \\ \text{such that} & \mathbf{a}_i^T \mathbf{x} + b_i = 0, i = 1, \dots, m_E, \\ & \mathbf{a}_i^T \mathbf{x} + b_i \leq 0, i = m_E + 1, \dots, m. \end{array}$$

Conic Linear Program A conic linear program is an optimization problem where the underlying space H is a general finite dimensional space, all the functions $f, g, i = 1, \dots, m$, are affine functions on H and the set in the geometric constraint is a specific set called convex cone.

Linear Integer Programming A linear integer programming problem is an optimization problem where the underlying space $H = \mathbb{R}^n$, all functions $f, g_i, i = 1, \dots, m$ are affine functions on $\mathbb{R}^n$ and the geometric constraint C is contained in the set of integers:

$$\begin{array}{ll} \min_{x \in \mathbb{R}^n} & f(\mathbf{x}) \\ \text{such that} & g_i(\mathbf{x}) = 0, i = 1, \dots, m_E, \\ & g_i(\mathbf{x}) \leq 0, i = m_E + 1, \dots, m, \\ & x \in C \subseteq \mathbb{Z}^n \end{array}$$

Chapter 2

Mathematical Background

2.1 Vectors and Matrices

These notes will skip some preliminaries that you should know from first and second year.

If $\mathbf{x}^T \mathbf{y} = 0$, then we say the two vectors $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.

Definition. (Vector Norm) A *vector norm* on $\mathbb{R}^n$ is a function $\|\cdot\|$ from $\mathbb{R}^n$ to $\mathbb{R}$ such that

1. $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}$.
2. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. (Triangle inequality)
3. $\|\alpha \mathbf{x}\| = |\alpha| \|\mathbf{x}\|$ for all $\alpha \in \mathbb{R}, \mathbf{x} \in \mathbb{R}^n$.

An $(n \times n)$ matrix A is said to be symmetric if $A = A^T$. We consider eigenvalues in non-ascending order.

Theorem. (Spectral Decomposition Theorem) Let $A \in \mathbb{R}^{n \times n}$ be a symmetric $n \times n$ matrix. Then, there exists an orthogonal matrix Q (that is, $Q^T Q = Q Q^T = I_n$) such that $A = Q D Q^T$ where $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $Q = [\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n]$ where $\mathbf{v}_i$ is an eigenvector of A corresponding to the eigenvalue λ_i .

For a symmetric $(n \times n)$ matrix A and any $\mathbf{x} \neq \mathbf{0}$, we define the so-called Rayleigh quotient as

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\|\mathbf{x}\|_2^2}.$$

Then, we have, for any $\mathbf{x} \neq \mathbf{0}$,

$$\lambda_{\min}(A) \leq R_A(\mathbf{x}) \leq \lambda_{\max}(A).$$

Theorem. (Variational Characterization for Extremal Eigenvalues) For a symmetric $(n \times n)$ matrix A ,

$$\lambda_{\max}(A) = \max\{R_A(\mathbf{x}) : \mathbf{x} \neq \mathbf{0}\} = \max_{\mathbf{x} \in \mathbb{R}^n} \{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\|_2 = 1\}$$

and

$$\lambda_{\min}(A) = \min\{R_A(\mathbf{x}) : \mathbf{x} \neq \mathbf{0}\} = \min_{\mathbf{x} \in \mathbb{R}^n} \{\mathbf{x}^T A \mathbf{x} : \|\mathbf{x}\|_2 = 1\}.$$

Definition. (Positive Definite Matrices) A matrix $A \in \mathbb{R}^{n \times n}$ is

- *positive definite* $\iff \mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0$
- *positive semi-definite* $\iff \mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *negative definite* $\iff \mathbf{x}^T A \mathbf{x} \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$
- *indefinite* $\iff$ there exists $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n : \mathbf{x}^T A \mathbf{x} > 0$ and $\mathbf{y}^T A \mathbf{y} < 0$

A symmetric matrix $A \in \mathbb{R}^{n \times n}$

- has n real eigenvalues $\lambda_i, i = 1, \dots, n$
- Determinant: $\det(A) = \prod_{i=1}^n \lambda_i$
- Trace: $\text{tr}(A) := \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$

Proposition. A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is

- *positive definite* $\iff \lambda_i > 0$ for all $i = 1, \dots, n$
- *positive semi-definite* $\iff \lambda_i \geq 0$ for all $i = 1, \dots, n$
- *negative definite* $\iff \lambda_i < 0$ for all $i = 1, \dots, n$
- *negative semi-definite* $\iff \lambda_i \leq 0$ for all $i = 1, \dots, n$
- *indefinite* $\iff$ there exists $i, j : \lambda_i > 0$ and $\lambda_j < 0$

Theorem. (Checking Positive (Semi-)definiteness) For any eigenvalues λ of a symmetric matrix $A \in \mathbb{R}^{n \times n}$, there exists $i_0 \in \{1, \dots, n\}$ such that

$$\lambda - A_{i_0 i_0} \leq \sum_{j \neq i_0} |A_{i_0 j}|.$$

Definition. (Diagonally Dominant) A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is called diagonally dominant if

$$|A_{ii}| \geq \sum_{j \neq i} |A_{ij}|, \text{ for each } i \in \{1, \dots, n\}.$$

In addition, if

$$|A_{ii}| > \sum_{j \neq i} |A_{ij}|, \text{ for each } i \in \{1, \dots, n\},$$

then we say A is strictly diagonally dominant.

Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Suppose that A is diagonally dominant and all of its diagonal elements are non-negative, then any eigenvalue of A must be non-negative, and so, A is positive semi-definite.

Definition. (Matrix Norm) A *matrix norm* on $\mathbb{R}^{p \times n}$ is a function $\|\cdot\|$ from $\mathbb{R}^{p \times n}$ to $\mathbb{R}$ such that

1. $\|X\| \geq 0$ for all $X \in \mathbb{R}^{p \times n}$ and $\|X\| = 0 \iff X = 0_{p \times n}$.
2. $\|X + Y\| \leq \|X\| + \|Y\|$ for all $X, Y \in \mathbb{R}^{p \times n}$. (Triangle inequality)
3. $\|\alpha X\| = |\alpha| \|X\|$ for all $\alpha \in \mathbb{R}, X \in \mathbb{R}^{p \times n}$.

One of the most widely used matrix norm is: **Frobenius norm**: $\|X\|_F = \left(\sum_{i=1}^p \sum_{j=1}^n X_{ij}^2 \right)^{\frac{1}{2}}$.

We can also define a dot product between two matrices $A, B \in \mathbb{R}^{p \times n}$ as

$$A \cdot B = \text{tr}(A^T B).$$

2.2 Local and Global Minima

Let H be a finite dimensional space and let Ω be a set in H . Let $f : H \rightarrow \mathbb{R}$ be a function on H and let $\|\cdot\|$ be a norm in H .

Definition. (Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$. The global minimum is $f(\mathbf{x}^*)$.

Definition. (Strict Global Minimum) A point $\mathbf{x}^* \in \Omega$ is a *strict global minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \Omega, \mathbf{x} \neq \mathbf{x}^*$.

Definition. (Local Minimum) A point $\mathbf{x}^* \in \Omega$ is a *local minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $\|\mathbf{x} - \mathbf{x}^*\| \leq \delta$. Then $f(\mathbf{x}^*)$ is a local minimum.

Definition. (Strict Local Minimum) A point $\mathbf{x}^* \in \Omega$ is a *strict local minimizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $0 < \|\mathbf{x} - \mathbf{x}^*\| \leq \delta$.

Definition. (Global Maximum) A point $\mathbf{x}^* \in \Omega$ is a *global maximizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) \geq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$. The global maximum is $f(\mathbf{x}^*)$.

Definition. (Strict Global Maximum) A point $\mathbf{x}^* \in \Omega$ is a *strict global maximizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if $f(\mathbf{x}^*) > f(\mathbf{x})$ for all $\mathbf{x} \in \Omega, \mathbf{x} \neq \mathbf{x}^*$.

Definition. (Local Maximum) A point $\mathbf{x}^* \in \Omega$ is a *local maximizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) \geq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $\|\mathbf{x} - \mathbf{x}^*\| \leq \delta$. Then $f(\mathbf{x}^*)$ is a local maximum.

Definition. (Strict Local Maximum) A point $\mathbf{x}^* \in \Omega$ is a *strict local maximizer* of $f(\mathbf{x})$ over $\Omega \subseteq H$ if and only if there exists a $\delta > 0$ such that $f(\mathbf{x}^*) > f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$ with $0 < \|\mathbf{x} - \mathbf{x}^*\| \leq \delta$.

Definition. (Closed Sets) A set Ω is said to be closed if it contains all the limits of convergent sequences of points in Ω .

Definition. (Bounded Sets) A set Ω is said to be bounded if there exists $R > 0$ such that $\Omega \subseteq \{x : \|x\| \leq R\}$.

Proposition (Existence of Global Solution: Type I) Let H be a finite dimensional space and let Ω be a closed and bounded set in H and let f be continuous on Ω . Then the global minima of f over Ω exist.

Proposition (Existence of Global Solution: Type II) Let H be a finite dimensional space and let $\|\cdot\|$ be a norm in H . Let Ω be a closed set in H and let f be continuous on Ω . Suppose that f is coercive in the sense that

$$\lim_{\|x\| \rightarrow \infty} f(x) = +\infty.$$

Then the global minima of f over Ω exist.

Relaxation Let H be a finite dimensional space. If $f : H \rightarrow \mathbb{R}$ and $\overline{\Omega} \subseteq \Omega$ then

$$\min_{x \in \Omega} f(\mathbf{x}) \leq \min_{\mathbf{x} \in \overline{\Omega}} f(\mathbf{x}).$$

Chapter 3

Solutions to Tutorial Problems

Note the problems are not available here. Please see course resources for reference.

3.1 Optimization Model Formulations

1. Decision Variables

- x_1 - exterior paints to produce (in tons)
- x_2 - interior paints to produce (in tons)

Objective Function

- $f(x_1, x_2) = 3000x_1 + 2000x_2$

Constraints

- $2x_1 + x_2 \leq 6$ (raw material A)
- $x_2 + 2x_1 \leq 8$ (raw material B)
- $x_2 \leq x_1 + 1$ (daily demand)
- $x_2 \leq 2$ (maximum demand)
- $x_1, x_2 \geq 0$ (hidden constraint)

So the problem can be formulated as,

$$\begin{aligned} & \max_{\mathbf{x}=(x_1, x_2) \in \mathbb{R}^2} && 3000x_1 + 2000x_2 \\ & \text{subject to} && 2x_1 + x_2 \leq 6 \\ & && x_2 + 2x_1 \leq 8 \\ & && x_2 \leq x_1 + 1 \\ & && x_2 \leq 2 \\ & && x_1, x_2 \geq 0 \end{aligned}$$

2. Decision Variables

- x_1 - quantity of corn (in kgs)
- x_2 - quantity of tankage (in kgs)
- x_3 - quantity of alfalfa (in kgs)

Objective Function

- $f(x_1, x_2, x_3) = 42x_1 + 36x_2 + 30x_3$

Constraints

- $90x_1 + 20x_2 + 40x_3 \geq 200$ (carbohydrates)
- $30x_1 + 80x_2 + 60x_3 \geq 180$ (proteins)
- $10x_1 + 20x_2 + 60x_3 \geq 150$ (vitamins)
- $x_1, x_2, x_3 \geq 0$ (hidden constraint)

So the problem can be formulated as,

$$\begin{aligned}
& \min_{\mathbf{x}=(x_1, x_2, x_3) \in \mathbb{R}^3} && 42x_1 + 36x_2 + 30x_3 \\
& \text{subject to} && 90x_1 + 20x_2 + 40x_3 \geq 200 \\
& && 30x_1 + 80x_2 + 60x_3 \geq 180 \\
& && 10x_1 + 20x_3 + 60x_3 \geq 150 \\
& && x_1, x_2, x_3 \geq 0
\end{aligned}$$

3.2 Mathematical Background

1. (a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Observe that $\lambda_1 + \lambda_2 = \text{tr}(A) = 1 + 4 > 0$ and $\lambda_1 \lambda_2 = \det(A) = 4 - 4 = 0$. Hence either $\lambda_1 = 5, \lambda_2 = 0$ or $\lambda_1 = 0, \lambda_2 = 5$ so $\lambda_1, \lambda_2 \geq 0$. Thus A is positive semi-definite.

(b) $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. Observe that $\lambda_1 \lambda_2 = \det(A) = 3 - 4 < 0$. Thus A is indefinite.

(c) $A = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & 4 \end{pmatrix}$. Observe that $|6| > |1| + |2|, |3| > |1| + |1|, |4| > |2| + |1|$, so A is strictly diagonal dominant. Furthermore, it has positive diagonal entries. Thus A is positive definite.

(d) $A = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 3 & 1 \\ 2 & 1 & -1 \end{pmatrix}$. Let $d_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $d_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Then, $d_1^T A d_1 = 6 > 0$ and $d_2^T A d_2 = -1$. Thus A is indefinite.

(e) $A = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 6 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. The eigenvalues of A are 2, λ_1 and λ_2 . The eigenvalues λ_1 and λ_2 are eigenvalues of the submatrix $A_0 = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$ which is positive definite as $\text{tr}(A_0) = 7 > \det(A_0) = 2 > 0$. Thus A is positive definite.

(f) $A = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}$. Observe that A is diagonal dominant with non-negative diagonal elements. Thus A is positive semi-definite.

2. (a) **HI**

(b) $\Omega = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$. Let $y, z \in \Omega$ and $\lambda \in [0, 1]$. Then

$$\begin{aligned}
\|\lambda y + (1 - \lambda)z\|_2 &\leq \|\lambda y\|_2 + \|(1 - \lambda)z\|_2 \\
&= \lambda\|y\|_2 + (1 - \lambda)\|z\|_2 \\
&< 1
\end{aligned}$$

since $y, z \in \Omega \implies \|y\|, \|z\| < 1$ and $\lambda \in [0, 1]$. Hence Ω is convex.

- (c) $\Omega = \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2 \leq 1\} \cup \{\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}$. Consider $y = (-1, 0)$ and $z = (0, 10)$ then $y, z \in \Omega$. However the midpoint $(-\frac{1}{2}, 5) \notin \Omega$. Hence Ω is not convex.

3. **HI**

4. (1) **HI**

(2) **HI**

(3) $f(x_1, x_2, x_3) = x_1^4 + x_2^4 + x_3^4 - \cos(x_1^2 + x_2^2 + x_3^2)$. Observe that as $\|\mathbf{x}\| \rightarrow \infty$, $x_1^4 + x_2^4 + x_3^4 \rightarrow \infty$ and $\cos(x_1^2 + x_2^2 + x_3^2) \leq 1$ for all $\mathbf{x}$. So their difference approaches ∞ . Hence f is coercive.

(4) $f(x_1, x_2) = e^{x_1^2} + e^{x_2^2} - x_1^2 - x_2^2$. We have

$$\begin{aligned} e^{x_1^2} + e^{x_2^2} - x_1^2 - x_2^2 &= e^{x_1^2} + e^{x_2^2} - (x_1^2 + x_2^2) \\ &= e^{x_1^2} + e^{x_2^2} - \|\mathbf{x}\|_2^2 \\ &\geq 2 \left(e^{x_1^2} e^{x_2^2} \right)^{\frac{1}{2}} - \|\mathbf{x}\|_2^2 && \text{(using } a + b \geq 2\sqrt{ab} \text{)} \\ &= 2 \left(e^{x_1^2 + x_2^2} \right)^{\frac{1}{2}} - \|\mathbf{x}\|_2^2 \\ &= 2e^{\frac{\|\mathbf{x}\|_2^2}{2}} - \|\mathbf{x}\|_2^2 \end{aligned}$$

. Then since $\lim_{\|\mathbf{x}\| \rightarrow \infty} \frac{e^{\frac{\|\mathbf{x}\|_2^2}{2}}}{\|\mathbf{x}\|_2^2} \rightarrow \infty$ then $f \rightarrow \infty$. Hence f is coercive.

5. **HI**

6. **HI**

7. (1) **HI**

(2) $f(X) = \text{rank}(X)$ for all $X \in S^n$. Let $X = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $Y = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Then $\text{rank}(X) = 1$ and $\text{rank}(Y) = 1$, but $\text{rank}(\frac{X+Y}{2}) = 2$. So $\text{rank}(\frac{X+Y}{2}) > \text{rank}(X) + \text{rank}(Y)$. Thus f is not convex.

(3) $f(X) = \det(X)$ for all $X \in S^n$. Similarly to above, $f(\frac{X+Y}{2}) = \frac{1}{4} > f(\frac{X}{2}) + f(\frac{Y}{2}) = 0 + 0$. Thus f is not convex.

3.3 Geometry of Linear Programs and Simplex Methods

1. (a) $\Omega_1 = \{\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1, x_i \geq 0, i = 1, 2, 3\}$. We see that $n = 3$ and $m = 1$, then hence an extreme point has $n - m = 2$ entries are zero. So candidates for extreme points are

$$\mathbf{x}_1^* = (1 \ 0 \ 0), \quad \mathbf{x}_2^* = (0 \ 1 \ 0), \quad \mathbf{x}_3^* = (0 \ 0 \ 1).$$

Now $x_1 + x_2 + x_3 = 1 \iff \mathbf{a}_1^T \mathbf{x} = b_1$ with $\mathbf{a}_1 = (1 \ 1 \ 1)^T$ and $b_1 = 1$. So $A = \mathbf{a}_1^T = (1 \ 1 \ 1)$. For $\mathbf{x}_1^*$, as only the first component is non-zero and $\{A_1\}$ is linearly independent then $\mathbf{x}_1^*$ is an extreme point of Ω_1 . Similarly $\mathbf{x}_2^*$ and $\mathbf{x}_3^*$ are also extreme points of Ω_1 .

- (b) $\Omega_2 = \{\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n : \sum_{i=1}^n x_i = 2, x_i \geq 0, i = 1, \dots, n\}$. Similarly $m = 1$, so there are $n - 1$ zero entries in the extreme points. Then from above, it follows that each $2\mathbf{e}_i = (0 \ \dots \ 2 \ \dots \ 0) \in \mathbb{R}^n$ is an extreme point of Ω_2 .

(c) $\Omega_3 = \{\mathbf{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 + x_3 = 1, x_2 + x_4 = 2, x_i \geq 0, i = 1, \dots, 4\}$. Note that $n = 4$ and $m = 2$, so extreme points have $n - m = 2$ entries are zero. Then note that

$$\begin{aligned} x_1 + x_2 + x_3 = 1 &\iff \mathbf{a}_1^T \mathbf{x} = 1 \\ x_2 + x_4 = 2 &\iff \mathbf{a}_2^T \mathbf{x} = 1 \end{aligned}$$

where $\mathbf{a}_1 = (1 \ 1 \ 1 \ 0)^T$ and $\mathbf{a}_2 = (0 \ 1 \ 0 \ 1)^T$. So, $A = \begin{pmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$.

- i. If $x_1 = x_2 = 0$, then $x_3 = 1, x_4 = 2$. As $\{A_3, A_4\}$ are linearly independent, $(0 \ 0 \ 1 \ 2)$ is an extreme point.
- ii. If $x_1 = x_3 = 0$, then $x_2 = 1, x_4 = 1$. As $\{A_2, A_4\}$ are linearly independent, $(0 \ 1 \ 0 \ 1)$ is an extreme point.
- iii. If $x_1 = x_4 = 0$, then $x_2 = 2$ and $x_3 = -1$ which is not in the feasible region.
- iv. If $x_2 = x_3 = 0$, then $x_1 = 1, x_4 = 2$. As $\{A_1, A_4\}$ are linearly independent, $(1 \ 0 \ 0 \ 2)$ is an extreme point.
- v. If $x_2 = x_4 = 0$, then $x_2 + x_4 \neq 2$ which is impossible.
- vi. If $x_3 = x_4 = 0$, then $x_2 = 2, x_1 = -1$ which is not in the feasible region.

Thus extreme points for Ω_3 are $\mathbf{x} = (0 \ 0 \ 1 \ 2), (0 \ 1 \ 0 \ 1), (1 \ 0 \ 0 \ 2)$.

2. HI

3. HI

4. Decision Variables

Constraints

To

- x_1 - number of inkjet printers
- x_2 - number of laser printers

- $x_1 + 2x_2 \leq 40$ (total hours)
- $3x_1 + x_2 \leq 45$ (storage space)
- $x_2 \leq 12$ (market capacity)
- $x_1, x_2 \geq 0$ (hidden constraint)

Objective Function

- $f(x_1, x_2) = 30x_1 + 25x_2$

So the problem can be formulated as,

In terms of standard form, we have

$$\begin{aligned} \max_{\mathbf{x}=(x_1, x_2) \in \mathbb{R}^2} \quad & 30x_1 + 25x_2 \\ \text{subject to} \quad & x_1 + 2x_2 \leq 40 \\ & 3x_1 + x_2 \leq 45 \\ & x_2 \leq 12 \\ & x_1, x_2 \geq 0. \end{aligned}$$

$$\begin{aligned} \min_{\mathbf{x}=(x_1, x_2) \in \mathbb{R}^2} \quad & -30x_1 - 25x_2 \\ \text{subject to} \quad & x_1 + 2x_2 + s_1 = 40 \\ & 3x_1 + x_2 + s_2 = 45 \\ & x_2 + s_3 = 12 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$

use simplex method, we first rewrite the above standard form as

$$\begin{aligned} \text{Minimise} \quad & z \\ \text{subject to} \quad & z + 30x_1 + 25x_2 = 0 \\ & x_1 + 2x_2 + s_1 = 40 \\ & 3x_1 + x_2 + s_2 = 45 \\ & x_2 + s_3 = 12 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	30	25	0	0	0	0	
s_1	0	1	2	1	0	0	40	non-basic variables: x_1, x_2
s_2	0	3	1	0	1	0	45	
s_3	0	0	1	0	0	1	12	

With our entering variable x_1 and outgoing variable s_2 :

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	0	15	0	-10	0	-450	
s_1	0	0	$\frac{5}{3}$	1	$-\frac{1}{3}$	0	25	non-basic variables: s_2, x_2
x_1	0	1	$\frac{1}{3}$	0	$\frac{1}{3}$	0	15	
s_3	0	0	1	0	0	1	12	

Now entering variable x_2 and outgoing variable s_3 :

Basic Variables	Z	x_1	x_2	s_1	s_2	s_3	b	
	1	0	0	0	-10	-15	-630	
s_1	0	0	0	1	$-\frac{1}{3}$	$-\frac{5}{3}$	5	non-basic variables: s_2, s_3
x_1	0	1	0	0	$\frac{1}{3}$	$-\frac{1}{3}$	11	
x_2	0	0	1	0	0	1	12	

Thus, an optimal solution is $x_1^* = 11, x_2^* = 12$.

3.4 Duality and Sensitivity Analysis for Linear Programs

3.5 Interior Point Method for Linear Programs and Applications for Linear Programs

3.6 Network Problems

3.7 Conic Programs

3.8 Integer Programs